Microcontroller PYQ 2019–2024 — Topic-wise Question Bank

Compiled from uploaded papers; grouped by syllabus chapter

Chapter 1: Peripheral Interfacing Fundamentals

1. What is meant by computer peripheral interfacing? Explain with example.

[Repeated 4 times: 2024-Q1a, 2023-Q1a, 2022-Q1a, 2021-Q1a]

2. Define computer peripheral interfacing with example.

[Repeated 2 times: 2020-Q1a, 2019-Q1a]

3. What do you mean by computer peripheral interfacing?

[1 time: 2021-Q1a]

4. Discuss peripheral-mapped I/O and memory-mapped I/O with necessary control signals and instruction.

[Repeated 2 times: 2024-Q1c, 2023-Q1b]

5. Distinguish between port-addressed I/O and memory-mapped I/O.

[Repeated 3 times: 2023-Q1b, 2022-Q1b, 2021-Q1b]

6. Compare port-addressed I/O and memory-mapped I/O on basis of address length, control signal, and instruction.

[1 time: 2020-Q1b]

7. How many I/O devices can be interfaced with port-addressed I/O? Explain.

[Repeated 2 times: 2023-Q1c, 2022-Q1c]

8. Is it possible in interfacing, IN 20H and then OUT 20H? Explain why or why not.

[Repeated 2 times: 2022-Q2a, 2021-Q1c]

9. Can it be LDA 8000H and then STA 8000H? Explain why or why not.

[1 time: 2021-Q1c]

10. In interfacing program, is it possible to write, IN 20H and then OUT 20H? Explain why or why not.

[Repeated 2 times: 2020-Q1c, 2019-Q1c]

11. Is it possible in interfacing, IN 40H and then OUT 40H? Explain why or why not.

[1 time: 2019-Q1c]

12. What do you know about the term 'MPU'? Is 8085 an MPU? Why or why not?

[1 time: 2019-Q2b]

13. Handshake signals are used to interface I/O devices with processors. Can you explain why and how these signals are used?

[Repeated 2 times: 2022-Q3a, 2019-Q2a]

14. What is meant by computer peripheral interfacing? Explain with example.

[1 time: 2023-Q1a]

Chapter 2: Address Decoding and I/O Mapping

1. Distinguish between absolute and partial address decoding.

[Repeated 3 times: 2024-Q1d, 2023-Q2b, 2022-Q2a]

2. Discuss absolute address decoding and partial address decoding with example.

[Repeated 2 times: 2023-Q2b, 2022-Q2a]

3. Define absolute address decoding and partial address decoding with example.

[1 time: 2021-Q2a]

4. What do you mean by address decoding? What are the needs of address decoding?

[1 time: 2024-Q1b]

5. Discuss how device select pulse is generated prior to transfer data in between processor and I/O devices.

[1 time: 2020-Q2b]

6. Discuss how to interface eight DIP Switches using 74LS138 Decoder and 74LS244 Tri-State Octal Buffer.

[Repeated 3 times: 2023-Q4a, 2020-Q2c, 2020-Q2c]

7. Draw the circuitry to interface eight DIP switches using a 74LS138 decoder and a 74LS244 buffer.

[1 time: 2020-Q2c]

Chapter 3: Timing Waveforms and Instruction Execution

1. Draw the timing-waveform of the execution of instruction OUT 01H.

[Repeated 3 times: 2024-Q2a, 2023-Q2a, 2021-Q2b]

2. Draw the timing-waveform of the execution of instruction OUT O1H.

[1 time: 2024-Q2a]

3. Draw the timing-waveform of the execution of instruction IN 84H.

[1 time: 2022-Q1c]

Chapter 4: Seven-Segment LED Interfacing

1. Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 5 to the output port.

[Repeated 2 times: 2024-Q2c, 2022-Q2c]

2. Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 7 to the output port.

[Repeated 3 times: 2023-Q2c, 2020-Q3a, 2020-Q2c]

3. Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 8 to the output port.

[1 time: 2021-Q2c]

Chapter 5: Safety Control and Appliance Interfacing

1. Discuss an interfacing circuit to design a safety control system for home appliances using a decoder, a buffer as an input port and a latch as an output port. Also write the program to operate the controlling system.

[Repeated 4 times: 2024-Q3a, 2023-Q3a, 2022-Q3b, 2021-Q3b]

2. Draw an interfacing circuit to design a safety control system for some home appliances using a decoder, a buffer and a latch. Also write the program to operate the controlling system.

[Repeated 2 times: 2020-Q3b, 2019-Q3b]

Chapter 6: Conveyor Belt / Process Monitoring Interfacing

1. Draw an interfacing circuit to monitor various processes (conveyer belts) on the floor of a manufacturing plant and write a program to perform the following operations: (i) Turn on the five conveyer belts according to the ON/OFF positions of switches S4–S0 at port F1H. (ii) Turn off the conveyer belts and turn on the emergency alarm only when both switches - S from port F1H and S7 from port F2H - are triggered. (iii) Monitor the switches continuously.

[1 time: 2024-Q3b]

Chapter 7: Programmable Peripheral Interface (PPI) and 8155

1. What do you mean by Programmable Peripheral Interfacing (PPI)?

[Repeated 3 times: 2024-Q4a, 2023-Q7a, 2022-Q6a]

2. What is meant by PPI?

[Repeated 3 times: 2021-Q6a, 2020-Q8a, 2019-Q7a]

3. Discuss the I/O Ports and Timer of 8155 with block diagram.

[Repeated 2 times: 2024-Q4b, 2023-Q8a]

4. Discuss the control word format of 8155.

[Repeated 3 times: 2023-Q7b, 2022-Q6b, 2021-Q7b]

5. Design a square-wave generator with a pulse width of 100μs by using the 8155 timer. Set up the timer in Mode 1 if the clock frequency is 3 MHz.

[Repeated 5 times: 2024-Q4c, 2023-Q7c, 2022-Q6c, 2021-Q7c, 2019-Q6b]

6. Is it possible to design a square-wave generator with a pulse width of 100μs by using the 8155 timers? If your answer is 'yes' then explain how? If your answer is 'No' then explain why.

[1 time: 2019-Q6b]

7. What do you know about the I/O Ports and Timer of 8155.

[1 time: 2019-Q6a]

Chapter 8: 8255A PPI

1. Discuss 8255A control word formats.

[Repeated 2 times: 2024-Q5a, 2023-Q8a]

2. Discuss BSR mode of 8255A with a suitable example.

[Repeated 3 times: 2024-Q5b, 2023-Q8a, 2022-Q8a]

3. Why BSR mode of 8255A is used? Give an example.

[Repeated 3 times: 2022-Q7a, 2021-Q7a, 2020-Q8d]

4. Interface 8255A ports in mode 0 with processor for the following conditions: (i) Use port B and port CL as input ports to read inputs from DIP switches and port A and port CU as output ports to display output at leads. (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port CL at port CU.

[Repeated 4 times: 2024-Q5c, 2023-Q8c, 2022-Q8b, 2021-Q8b]

5. Discuss how data is transferred bidirectionally between two computers using port A of 8255A in Mode 2.

[1 time: 2021-Q8a]

6. What is key debounce? How many ways it can be generated? Explain.

[1 time: 2021-Q8b]

7. How a 20-key matrix keyboard is interfaced with processor using an 8255A PPI.

[1 time: 2021-Q8c]

8. What is Control Word? Determine the control word for the following configuration of 8255:- Port A - Output, Mode of port A - Mode 1, Port B - Output, Mode of port B - Mode 0, Port C lower (pins PC0 - PC2) - Output.

[1 time: 2020-Q7b]

9. Use 8255A in mode 1 where a printer is interfaced using port B with status I/O. Write initialization instructions and a printer subroutine to output characters those are stored in memory.

[1 time: 2020-Q7c]

10. What is PPI? Write down the 8255A PPI control word format for BSR mode.

[1 time: 2020-Q7a]

11. Interface 8255A ports in mode 0 with processor for the following conditions: (i) Use port B and port CL as input ports to read inputs from DIP switches and port A and port CU as output ports to display output at LEDs. (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port CL at port CU.

[Repeated 2 times: 2020-Q8c, 2019-Q8b]

Chapter 9: 8254 Programmable Interval Timer

1. Draw and discuss the block diagram of 8254 programmable interval timer.

[Repeated 3 times: 2022-Q7c, 2021-Q7c, 2020-Q8c]

2. Briefly discuss how a counter of 8254 is initialized and the value of the counter is read during counting.

[1 time: 2020-Q8c]

Chapter 10: DMA Controller (8257)

1. Describe the functional diagram of 8257 DMA controller.

[1 time: 2020-Q7a]

Chapter 11: Interrupts (8085)

1. Discuss the 8085 vectored interrupts with necessary diagram.

[Repeated 3 times: 2023-Q5c, 2022-Q4b, 2021-Q4b]

2. What do you mean by maskable and nonmaskable interrupt of 8085?

[Repeated 4 times: 2024-Q5a, 2022-Q5a, 2021-Q4a, 2019-Q4a]

3. Can the microprocessor be interrupted again before the completion of the first interrupt service routine? Explain.

[Repeated 5 times: 2024-Q5b, 2023-Q5b, 2021-Q4b, 2020-Q5d, 2020-Q5e]

4. Is there a minimum pulse width required for the INTR signal? Explain.

[Repeated 3 times: 2022-Q4c, 2021-Q4c, 2019-Q4b]

5. How long can the INTR pulse stay high?

[1 time: 2022-Q4a]

6. Explain the use of SIM and RIM instructions.

[Repeated 2 times: 2022-Q4c, 2019-Q5a]

7. Why SIM and RIM instructions are used?

[1 time: 2019-Q5a]

8. What is NMI? Mention the difference between RST 7.5 and RST 7.

[1 time: 2019-Q5b]

9. Write a main program to count continuously in binary with a one-second delay between each count and write a service routine to flash FFH six times when the program is interrupted, with some appropriate delay between each flash.

[1 time: 2023-Q4d]

10. Write a main program to count continuously in binary with a one-second delay between each count and write a service routine at XX70 to flash FFH five times when the program is interrupted, with some appropriate delay between each flash.

[Repeated 2 times: 2020-Q5b, 2019-Q4c]

11. Write a main program to count continuously in binary with a one-second delay between each count and write a service routine to flash FFH six times when the program is interrupted, with some appropriate delay between each flash.

[1 time: 2021-Q4d]

12. What is vectored interrupt? Discuss the vectored interrupts of 8085 processor.

[1 time: 2021-Q5a]

13. Write the difference between vectored and non-vectored interrupts.

[1 time: 2020-Q5c]

14. Define maskable, nonmaskable, vectored, and non-vectored interrupts of 8085.

[1 time: 2019-Q4a]

15. Design a 1-minute timer using a 60Hz power line as an interrupting source. The output ports display minutes and seconds in BCD. At the end of the minute, the output ports should continue displaying one minute and zero seconds.

[Repeated 2 times: 2021-Q5b, 2019-Q5c]

Chapter 12: A/D and D/A Interfacing

1. Discuss how an 8-bit A/D converter is interfaced with processor using status check I/O technique.

[Repeated 4 times: 2024-Q6a, 2023-Q6a, 2021-Q6a, 2020-Q6a]

2. Design an output port with the address FFH to interface the 1408 D/A converter that is calibrated for 0 to 10V range. Write a program to generate a continuous ramp waveform. Explain the operation of the 1408 for the calibration of a bipolar range +5V to -5V. Calculate the output Vo if the input is 100000002.

[Repeated 4 times: 2024-Q6b, 2023-Q6b, 2021-Q6b, 2020-Q6b]

3. Design an output port with the address FFH to interface the 1408 D/A converter that is calibrated for 0 to 5V range. Write a program to generate a continuous ramp waveform.

[1 time: 2021-Q6b]

4. Design an output port with the address FFH to interface the 1408 D/A converter that is calibrated for 0 to 10V range. Write a program to generate a continuous ramp waveform. Explain the operation of the 1408 for the calibration of a bipolar range +5V to -5V. Calculate the output Vo if the input is 1000000.

[1 time: 2022-Q5b]

Chapter 13: Time Delay Calculation

1. Discuss how time delay is calculated using register pair technique.

[Repeated 2 times: 2023-Q3a, 2022-Q3a]

2. Discuss how maximum time delay is calculated in 8085.

[Repeated 2 times: 2019-Q3a, 2019-Q4b]

3. Discuss how maximum time delay is calculated in 8085 using a register pair with a nested loop.

[1 time: 2022-Q3b]

4. Discuss how time delay is calculated using register pair technique with a counter value of FFFFH.

[1 time: 2021-Q3a]

5. Describe the delay calculation process using register pair.

[1 time: 2020-Q4a]

Chapter 14: ARM Cortex-M Register Set and Instructions

1. Explain PUSH and POP instruction of Cortex M3 or M4 with examples.

[1 time: 2024-Q1a]

2. Explain LDR and STR Instructions of Cortex M3 or M4 with examples.

[1 time: 2024-Q1b]

3. Explain MOVT Instruction of Cortex M3 or M4 with examples.

[1 time: 2024-Q1c]

4. Draw the ARM register set organization and explain the role of each category of registers.

[1 time: 2024-Q6a]

5. Differentiate between register R13 (SP) and register R14 (LR) in ARM architecture in terms of their working principle.

[1 time: 2024-Q6b]

Chapter 15: STM32 Register-Level Programming

1. Load the roll numbers of team leaders of five teams of your class into register R0 to R4. Write a register-level program for a STM32F103X or STM32F446X, (i) to discard the session and hall code from the roll number (ii) move the roll numbers stored in R0~R4 to a SRAM area, starting address is 0x2FF20F00.

[1 time: 2024-Q2a]

2. Assume there are 100 integer numbers stored in SRAM area, starting address is 0x2FF20000, with each integer having a value ranging from 0 to 255. Write a register-level program for STM32F103X or STM32F446X to sort these numbers in ascending order using Selection sort.

[1 time: 2024-Q2b]

3. Load the roll numbers of team leaders of five teams of your class into register R0 to R4. Write a register-level program for STM32F103X or STM32F446X, (i) to move the roll numbers stored in R0~R4 to SRAM area, starting address is 0x2FF20400, (ii) then move all roll numbers to SRAM area, starting address is 0x2FF20500.

[1 time: 2024-Q3a]

4. Store the ASCII codes of uppercase-letters into SRAM area staring from 0x2FF10000. Write a register-level program for STM32F103X or STM32F446X to convert uppercase-letters into corresponding lowercase-letters and store their ASCII codes into SRAM area, starting from 0x2FF20000.

[1 time: 2024-Q3b]

5. You have given four sets of positive integer number that are stored in the SRAM area, the

starting addresses and number of integers for each set are specified as follows: Set 1: Starting at address 0x2FF20000, contains 18 integers. Set 2: Starting at address 0x2FF20600, contains 09 integers. Set 3: Starting at address 0x2FF20900, contains 21 integers. Set 4: Starting at address 0x2FF20A00, contains 06 integers. Write a program for STM32 Cortex processor to determine the largest and smallest values from each set and store these values sequentially in memory.

[1 time: 2024-Q4]

Chapter 16: ARM Cortex-M4 Architecture and FPU

1. Draw the block diagram of ARM Cortex M4 and describe its major component.

[1 time: 2024-Q5a]

2. What are the main features of Floating Point Unit (FPU) of Cortex M4.

[1 time: 2024-Q5b]

Chapter 17: I²C Protocol

1. Explain the role of SDA and SCL lines in I2C communication protocol.

[1 time: 2024-Q6a]

2. What are the advantages and disadvantages of using I2C over SPI protocol?

[1 time: 2024-Q6b]

3. Describes the different fields of Message of I2C protocol.

[1 time: 2024-Q6c]

4. Explain why pull-up resistors are needed in I²C lines.

[1 time: 2024-Q8c]

Chapter 18: SPI Protocol

1. Describe the functions of MOSI, MISO, SCLK, and SS on SPI protocol.

[1 time: 2024-Q7a]

2. Illustrate with block diagram, how multiple Slaves could communicate with Master, when master has a single SS/CS line and has multiple separate SS/CS lines.

[1 time: 2024-Q7b]

3. Explain how full-duplex communication can be performed in SPI protocols.

[1 time: 2024-Q7c]

4. Suppose, two microcontrollers use the SPI communication protocol. If the clock frequency is 4 MHz, how long will it take to transmit 32 bits of data?

[1 time: 2024-Q8b]

Chapter 19: UART Protocol

1. Describe the structure of a UART data frame. Describe details of start bit, data bits, parity bit, and stop bits.

[1 time: 2024-Q8a]

2. Explain the steps of how flow control is managed in UART communication protocol.

[1 time: 2024-Q8b]

3. What are the roles of 12th bit of UART_CR1; 6th and 7th bits of USART_SR of STM32F446?

[1 time: 2024-Q8c]

4. If the baud rate is 9600 bps, how many characters can be transmitted in one second (as-

suming 8 data bits, 1 start, 1 stop)? Choose proper communication protocol for this purpose.

[1 time: 2024-Q7b]

Chapter 20: Microcontroller-to-Microcontroller Communication

1. Suppose you want to send the message "CSE" from one microcontroller to another microcontroller. You must confirm receipt of the data byte to the sender. The sender address is (130)10 and the receiver address is (130)10. Which communication protocol would be most suitable for this purpose. Design the packet format for that protocol and describe the protocol step by step with proper justification and illustration.

[1 time: 2024-Q7a]

2. Suppose you want to connect three microcontrollers for data exchange. Which protocol (UART, SPI, or I²C) would you choose and why? Describe the protocol with proper justification.

[1 time: 2024-Q8a]

Chapter 21: Microprocessor Architecture and Instruction Handling

1. Discuss how does a microprocessor differentiate between data (operand) and instruction code (opcode).

[1 time: 2023-Q4b]

2. Discuss with example how CALL and RET instructions work in between a main program and a subroutine.

[1 time: 2023-Q4c]

Chapter 22: Memory Addressability and Address Lines

1. How many memory locations can be addressed by a microprocessor with 14 address lines? How many address lines are necessary to address two megabytes (2048K) of memory?

[1 time: 2024-Q2b]

Total questions: 110
Chapters: 22
Most repeated question: "Design a square-wave generator with a pulse width of 100µs by using the 8155 timer..." — 5 times
Second most repeated: "Can the microprocessor be interrupted again before the completion of the first interrupt service routine?" — 5 times
Third most repeated: "Interface 8255A ports in mode 0..." — 4 times
Fourth most repeated: "Discuss an interfacing circuit to design a safety control system..." — 4 times
Fifth most repeated: "Discuss how an 8-bit A/D converter is interfaced..." — 4 times
Sixth most repeated: "Design an output port with the address FFH to interface the 1408 D/A converter..." — 4 times